	Baseline AUROC	Heritage-Informed AUROC
Seed 40	0.700	0.714
Seed 41	0.700	0.715
Seed 42	0.701	0.715
Seed 43	0.701	0.715
Seed 44	0.702	0.715

Table 4: Performance by model and seed for the two MDD prediction models fine-tuned on data from all five administrative regions.

Appendix

Data Pre-Processing for MDD Prediction

Our dataset included both numerical and categorical variables. Numerical features were standardized to control for differences in scale. Categorical variables were converted into numerical representations using periodic encoding, as required for tabular transformer models (Gorishniy, Rubachev, and Babenko 2022).

Tabular Transformer Model and Training for MDD Prediction

TF-Transformer was trained for the binary classification of individuals according to MDD diagnoses using each of the eight datasets for both models (Gorishniy et al. 2021; Huang et al. 2020). Our model was initialized with eight attention heads per block and three transformer blocks (or layers). The dropout rate for attention and dense layers was set to 0.2. We used 16 embedding dimensions to encode the variables in the model. The TF-Transformer model is known to be less sensitive to hyperparameter tuning (Gorishniy et al. 2021). Accordingly, we implemented a learning rate warm-up with linear decay (2e-5 to 1e-4) with other key hyper-parameters, including epochs (100), early stopping (5 epochs), weight decay (1e-4), batch size (512), and seed (42). The model was evaluated on the validation set using binary cross-entropy.

Model Performance with Multiple Seeds

Additional analyses were conducted to explore the effects of seed on model performance when using the Heritage-Informed versus the Baseline model. In doing so, we aimed to better understand how often and to what extent the Heritage-Informed model outperforms the Baseline model when both are trained and evaluated on the nationwide data. Both models were evaluated on identical data splits using 5 random seeds (40-44). AUROC differences were computed per seed and analyzed using a paired design. The Heritage-Informed model achieved consistently higher AUROC across seeds (mean difference = 0.014, see Table 4). A paired Wilcoxon signed-rank test on within-seed AUROC differences indicated an improvement approaching significance ($p = 0.06$). These results suggest that performance differences are attributable to variable selection rather than random seed effects. Based on the results discussed earlier, we argue these differences in performance may exist more clearly for some subgroups than others.

Feature Importance for MDD Prediction

We utilized the model weights of the transformer models after training to assess feature importance when evaluated on the test set. To do so, we calculate the local importances (P_i) using equation 1. This determines which features most contribute to the model predictions for each row of the test set. To summarize, equation 1 takes the sum of the attention scores from *h*-th head's attention map for a [CLS] token from the forward pass of the *l*-th layer per *i*-th sample (Gorishniy et al. 2021). This is then divided by the number of heads and layers (i.e., the depth of the model). The global importances (P) are calculated using equation 2, wherein the local importances (P_i) are averaged across all rows.

$$P_i = \frac{1}{n_{heads} \times l} \sum_{h,l} P_{ihl} \tag{1}$$

$$P = \frac{1}{n_{samples}} \sum_{i} P_i \tag{2}$$

	Overall, N = 363,643	Healthy Controls, N = 195,419 (54%)	MDD Group, N = 168,224 (46%)
Age (Years)	41.6 (13.1)	43.1 (12.9)	39.9 (13.0)
Sex			
Female	198,541 (55%)	95,945 (49%)	102,596 (61%)
Male	165,028 (45%)	99,460 (51%)	65,568 (39%)
Changed Sex	74 (<0.1%)	14 (<0.1%)	60 (<0.1%)
LGBT Identity			
LGBT	1,460 (0.4%)	637 (0.3%)	823 (0.5%)
Not LGBT	362,183 (99.6%)	194,782 (99.7%)	167,401 (99.5%)
Civil Status			
Married	162,528 (45%)	101,635 (52%)	60,893 (36%)
Not married	145,769 (40%)	70,085 (36%)	75,684 (45%)
Divorced	48,553 (13%)	20,380 (10%)	28,173 (17%)
Widowed	6,793 (1.9%)	3,319 (1.7%)	3,474 (2.1%)
Region			
Capital Region	112,791 (31%)	60,664 (31%)	52,127 (31%)
Middle Jutland	83,331 (23%)	44,026 (23%)	39,305 (23%)
North Jutland	33,105 (9.1%)	20,736 (11%)	12,369 (7.4%)
South Denmark	82,226 (23%)	41,280 (21%)	40,946 (24%)
Zealand	52,190 (14%)	28,713 (15%)	23,477 (14%)
Immigration Status			
Danish origin	329,132 (91%)	179,984 (92%)	149,148 (89%)
Descendant	3,580 (1.0%)	1,946 (1.0%)	1,634 (1.0%)
Immigrant	30,931 (8.5%)	13,489 (6.9%)	17,442 (10%)
Heritage 1			
Danish	329,226 (91%)	180,018 (92%)	149,208 (89%)
Non Western Immigrant	25,586 (7.0%)	10,641 (5.4%)	14,945 (8.9%)
Western Immigrant	8,770 (2.4%)	4,740 (2.4%)	4,030 (2.4%)
Unknown	61 (<0.1%)	20 (<0.1%)	41 (<0.1%)
Heritage 2			
Australia and New Zealand	61 (<0.1%)	34 (<0.1%)	27 (<0.1%)
Caribbean	57 (<0.1%)	29 (<0.1%)	28 (<0.1%)
Central America	81 (<0.1%)	42 (<0.1%)	39 (<0.1%)
Central Asia	27 (<0.1%)	8 (<0.1%)	19 (<0.1%)
Eastern Africa	1,210 (0.3%)	623 (0.3%)	587 (0.3%)
Eastern Asia	415 (0.1%)	263 (0.1%)	152 (<0.1%)
Eastern Europe	4,391 (1.2%)	2,051 (1.0%)	2,340 (1.4%)
Middle Africa	72 (<0.1%)	42 (<0.1%)	30 (<0.1%)
Northern Africa	1,135 (0.3%)	483 (0.2%)	652 (0.4%)
Northern America	374 (0.1%)	195 (<0.1%)	179 (0.1%)
Northern Europe	332,616 (91%)	181,806 (93%)	150,810 (90%)
South-eastern Asia	1,906 (0.5%)	1,190 (0.6%)	716 (0.4%)
South America	504 (0.1%)	238 (0.1%)	266 (0.2%)
Southern Africa	56 (<0.1%)	31 (<0.1%)	25 (<0.1%)
Southern Asia	5,060 (1.4%)	2,091 (1.1%)	2,969 (1.8%)
Southern Europe	3,417 (0.9%)	1,318 (0.7%)	2,099 (1.2%)
Unknown	70 (<0.1%)	24 (<0.1%)	46 (<0.1%)
Western Africa	271 (<0.1%)	160 (<0.1%)	111 (<0.1%)
Western Asia	9,770 (2.7%)	3,549 (1.8%)	6,221 (3.7%)
Western Europe	2,150 (0.6%)	1,242 (0.6%)	908 (0.5%)

Mean (SD); n (%)

Table 5: Sociodemographic characteristics of the matched cohort dataset used in our analyses.

	Overall, N = 363,643	Healthy Controls, N = 195,419 (54%)	MDD Group, N = 168,224 (46%)
Address Changes	0.3 (0.6)	0.2 (0.5)	0.3 (0.7)
Cohabitation Status			
Living alone	109,591 (30%)	43,917 (22%)	65,674 (39%)
Not living alone	254,052 (70%)	151,502 (78%)	102,550 (61%)
Highest Education			
Short education	76,950 (21%)	36,509 (19%)	40,441 (24%)
Medium education	165,799 (46%)	88,468 (45%)	77,331 (46%)
Long education	120,490 (33%)	70,304 (36%)	50,186 (30%)
Unknown	404 (0.1%)	138 (<0.1%)	266 (0.2%)
Income Rank			
1 (lowest)	79,457 (22%)	28,737 (15%)	50,720 (30%)
2	80,567 (22%)	35,674 (18%)	44,893 (27%)
3	96,241 (26%)	55,870 (29%)	40,371 (24%)
4 (highest)	107,378 (30%)	75,138 (38%)	32,240 (19%)
Country of Origin Economic Level			
Low income	2,636 (0.7%)	1,035 (0.5%)	1,601 (1.0%)
Lower middle income	5,277 (1.5%)	2,870 (1.5%)	2,407 (1.4%)
Upper middle income	15,451 (4.2%)	5,897 (3.0%)	9,554 (5.7%)
High income	338,388 (93%)	184,879 (95%)	153,509 (91%)
Unknown	1,891 (0.5%)	738 (0.4%)	1,153 (0.7%)

Mean (SD); n (%)

Table 6: Socioeconomic characteristics of the matched cohort dataset used in our analyses.

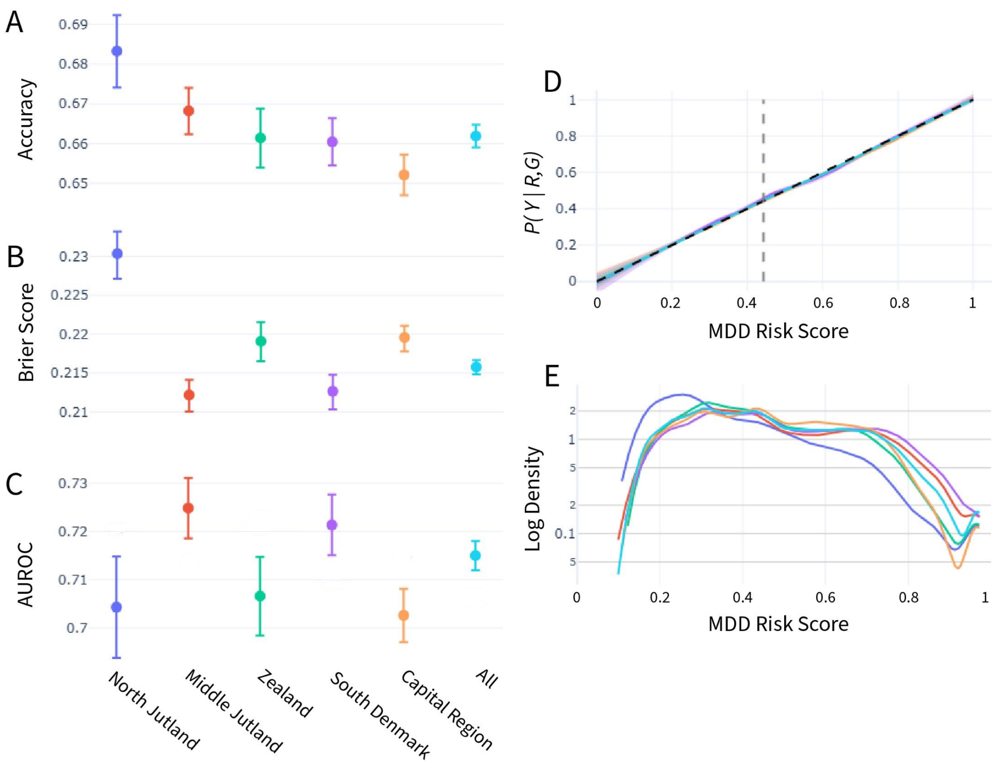

Figure 2: Fairness evaluation of the Heritage-Informed Model trained on data from the 5 administrative regions of Denmark, produced by the meval package (Sutariya and Petersen 2025). A. Accuracy score. B. Brier score. C. AUROC score. D. Reliability diagram (i.e., calibration curves) obtained using loess smoothing (Austin and Steyerberg 2014). E. MDD prediction density diagram. 95% confidence intervals obtained using DeLong's method (AUROC), the Wilson score interval (Acc), and percentile bootstrapping (Brier Score). Refer to (Sutariya and Petersen 2025) for more details.

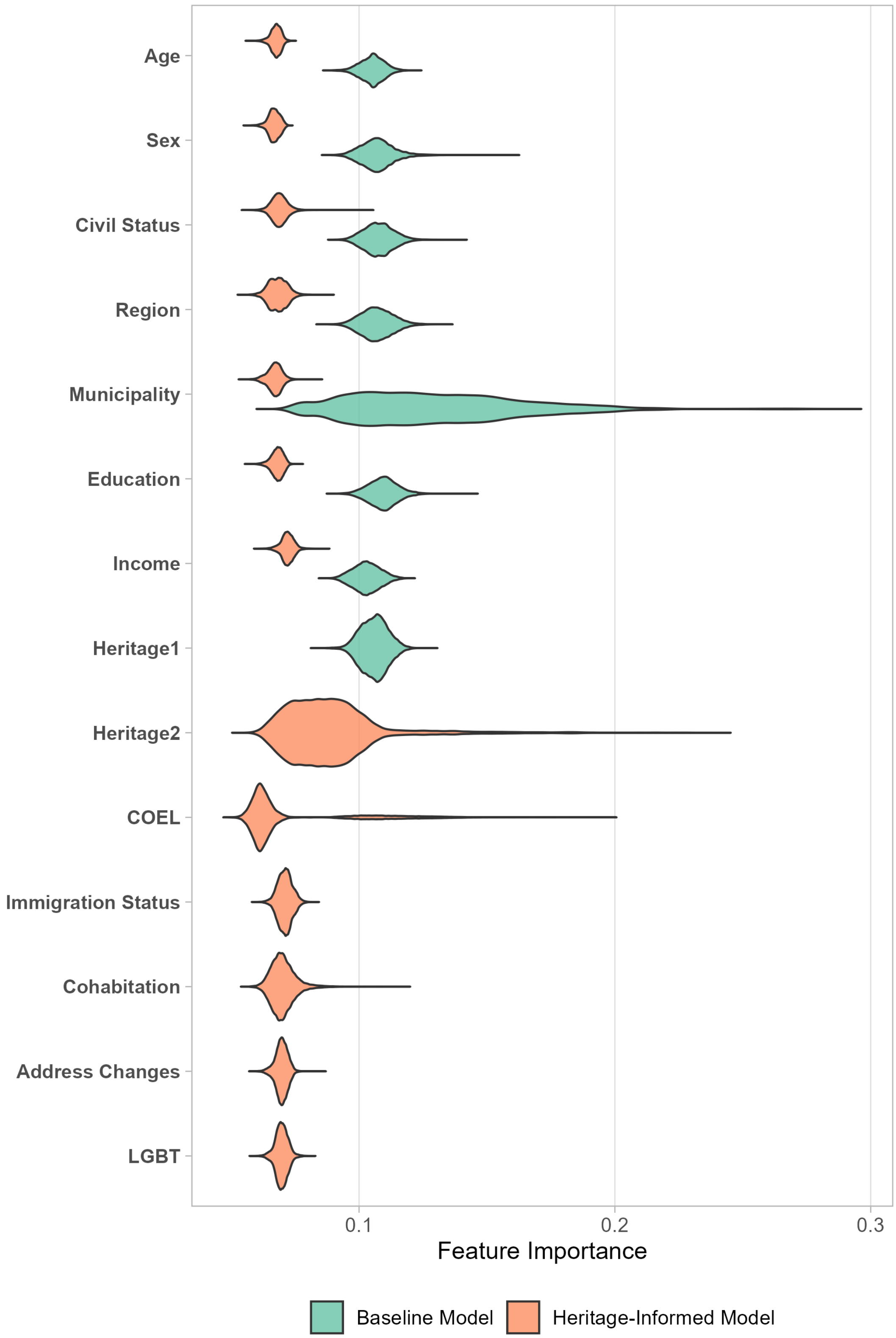

Figure 3: Global Feature Importance for the Baseline and the Heritage-Informed Models.